

Luis A. Rodriguez

Senior Software Engineer

Skills

Languages	C, JavaScript, TypeScript, CSS, HTML, Java
System and low level	Emscripten, Markdown, WebAssembly, Java Native Interface, Hardware Emulation, WebSerial, WebUSB, USB Communication, Cross-Compilation
Mobile/Web/Frameworks	Angular, Spring Boot, Lua, Gulp
Leadership	Test-Driven Development, Technical Leadership, Agile/Scrum, Project Coordination, Git

Work Experience

Texas Instruments -- Software Architect (2010-2025)

- **TI Connect CE:** Delivered cross-platform compatibility for a proprietary CARS communication protocol by porting the codebase to generate platform-native libraries (Windows/macOS). Established the USB communication architecture using an abstraction layer and integrated the functionality into the application via Java Native Interface (JNI).
- **Confidential Project: Device Emulation:** Designed and implemented the glue layer for targetting embedded code on the web by transpiling to WebAssembly. Coordinated crossfunctional team that did development and test for new features. Designed and implemented an automation entry point using WebSockets.
- **Examcalcs / TI 84 Plus CE ExamCalc:** Implemented emulators for ez80-based devices which have gone through our client's pilots involving 800k users without any reported issues. / Adapted z80 opcodes and emulated the TI-84 Silver Edition and TI-84 Plus C's ASICs. Ported Toshiba T4 cpu emulation from Java and implemented the TI-108 and TI-30MV's ASICs.
- **Lua Glue / Image Question type:** Designed and implemented the infrastructure for creating new classroom activities using the Lua scripting language. The C counterpart for the Lua bindings, as well as the Lua engine itself gets cross-compiled to run on different platform including iOS, Nucleus and desktop operating systems.
- **Examcalc Accessibiliy:** Implemented code needed in both z80 assembly and javascript to support market needs to make calculator emulators interact correctly with a screen reader for the vision impaired.
- **Ui And Background Thread Refactor:** Worked in conjunction with the development team to refactor the usage of JNI, improving performance and memory usage.
- **Open Source Auditing:** Responsible for reviewing software third party library lists to determine their eligibility for integrating into our products.
- **Web Apps Server-Side:** Led team in determining server configuration changes needed to address concerns brought up by IT security scans.

Digital Chocolate -- Game Engineering Manager (2009-2010)

- **Game Engineering Manager:** Coordinating project planning for various carriers in the US, working in conjunction with the sales and marketing teams to reduce production costs and improve revenue. Lead

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Location: Dallas, TX

and mentor the engineering team, offering tutorials and code deep-dives where needed.

- **Brew In App Purchase Library:** Created client module for Qualcomm's Application Value Billing API (C/C++) in conjunction with the team in Bangalore, India.
 - **3d Programmer (Gameloft):** In charge of porting multiple game titles for resource-limited, but 3d-capable devices. All ports used Java/J2ME
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Education

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

Completed coursework towards: Master's in Electronics and Telecommunications with specialization in control systems and automation, 2002-2005

Academic Project

Robot Control Over Ethernet

2004

Used RTLinux kernel modules on 2 computers to write a simple PD-controller in C with a GUI in Java on one computer to send the control signals over a TCP socket to another computer that interfaced with a 2-degree-of-freedom robot